Immigration and Credit in America

J. Anthony Cookson[1], Benedict Guttman-Kenney[2], & William Mullins[3]

[1]University of Colorado Boulder, [2]Rice University, [3]University of California San Diego

NBER Immigrants and the U.S. Economy, 6th March 2026

Motivation

- Immigrants to the U.S. bring their labor market experiences (e.g., education records, work history) helping them to navigate U.S. labor markets

Motivation

- Immigrants to the U.S. bring their labor market experiences (e.g., education records, work history) helping them to navigate U.S. labor markets
- Immigrants start life in the U.S. *without* a U.S. credit report
  - Any foreign credit history generally unobserved to U.S. lenders

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Data

- Representative anonymized sample of U.S. consumer credit reporting data, 2000 to 2024
- Data **matched** with information on whether a consumer is an **immigrant** & **year of immigration**, using sequential Social Security Number (SSNs) assignment

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Data

- Representative anonymized sample of U.S. consumer credit reporting data, 2000 to 2024
- Data **matched** with information on whether a consumer is an **immigrant** & **year of immigration**, using sequential Social Security Number (SSNs) assignment

What We Do

- Describe **selection** of Immigrants compared to Non-Immigrants in their creditworthiness
- Describe **assimilation** in creditworthiness & access to credit over life cycle
- How **dynamics** of life cycle outcomes vary by one year difference in age of immigration, (with fixed effects for birth year & ZIP5 geography)

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Data

- U.S. consumer credit reporting data matched with immigration data

What We Do

- Describe **selection** of Immigrants & **assimilation** dynamics over life cycle
  - Immigrants vs. same-age Non-Immigrants & Immigrants arriving at different ages

Immigrants (vs. Non-Immigrants), Immigrants Arriving At Older (vs. Younger) Ages:

- **Positive selection:** More creditworthy (once credit scored)

- Despite better creditworthiness, **lower long-term access** to auto loans & mortgages

Timing of immigration best explains most of our results **("history dependence")**

Our Paper Contributes To Labor Economics & Household Finance Literatures

- Prior work shows the large positive contribution of legal immigrants to U.S. economy (e.g., Azoulay et al., 2022; Bernstein et al., 2025)
- We show the **selection** of immigrants based on their credit behaviors, complementing work focusing on labor market outcomes (e.g., Borjas, 1985; Abramitzsky & Boustan, 2017)
- We show **assimilation** into the U.S. credit markets over life cycle, complementing work on labor & cultural assimilation (e.g., Borjas, 1985; Abramitzsky et al., 2014, 2020, 2021; Lubotsky, 2007; Bleakley & Chin, 2010; Doran, Geber, & Isen, 2022; Bailey et al., 2022)
- We contribute to the household finance literature that has largely not studied immigrants (e.g., Dobbie et al., 2021; Zillessen, 2022; Bakker et al., 2025; Gorback & Schubert, 2025)

Roadmap For Presentation

1. Data & Institutional Details
2. Credit Market Entry
3. Creditworthiness
4. Credit Access
5. Mechanisms

Data & Institutional Details

Data

University of Chicago Booth TransUnion Consumer Credit Panel (BTCCP)

- Anonymized data sample of 1 in 10 consumers with U.S. credit reports
- Monthly data from 2000 to 2024
- Individual tradeline data (e.g., details on each account, including opening dates) & consumer-level data (e.g., birth date, credit score, ZIP code)

Gibbs, Guttman-Kenney, Lee, Nelson, van der Klaauw & Wang (2025)

Data

University of Chicago Booth TransUnion Consumer Credit Panel (BTCCP)

- Anonymized data sample of 1 in 10 consumers with U.S. credit reports
- Monthly data from 2000 to 2024
- Individual tradeline data (e.g., details on each account, including opening dates) & consumer-level data (e.g., birth date, credit score, ZIP code)

Gibbs, Guttman-Kenney, Lee, Nelson, van der Klaauw & Wang (2025)

American Community Survey (ACS), 5-Year Waves (2006 to 2023)

- Validate immigration classification & evaluate mechanisms

Survey of Consumer Finances (SCF), 2022

- Evaluate mechanisms

Classifying Immigrants In The BTCCP

Before 2012, U.S. Social Security Administration assigned first 5-digits of 9-digit Social Security Numbers (SSNs) sequentially within states (Puckett, 2009)

Consumers assigned SSNs as adults are very likely to be (legal) immigrants

(Klopfer & Miller, 2024; Yonker, 2017; Doran, Gelber, & Isen, 2022; Bernstein et al., 2025)

We calculate:

$$\textit{SSN Age} = \textit{SSN Year} - \textit{Birth Year}$$

“Immigrants”: assigned SSN at age 21+. **“Non-Immigrants”:** assigned SSN at age < 21

Classifying Immigrants In The BTCCP

Before 2012, U.S. Social Security Administration assigned first 5-digits of 9-digit Social Security Numbers (SSNs) sequentially within states (Puckett, 2009)

Consumers assigned SSNs as adults are very likely to be (legal) immigrants

(Klopfer & Miller, 2024; Yonker, 2017; Doran, Gelber, & Isen, 2022; Bernstein et al., 2025)

We calculate:

$$\textit{SSN Age} = \textit{SSN Year} - \textit{Birth Year}$$

"Immigrants": assigned SSN at age 21+. **"Non-Immigrants":** assigned SSN at age < 21

Restrict To Birth Years 1975 to 1987 With SSN Ages 0 to 29:

6,299,287 consumers – 384,396 (6.1%) "Immigrants"

Follow **credit outcomes** in BTCCP for **ages 18 to 40**

Institutional Details On Lending In America

- **Information frictions** may reduce immigrant credit access:
  1. Immigrants have **no credit report on arrival**
     - **Lender's prior:** Consumer with no U.S. credit report is **high credit risk**
  2. After entering the credit system, immigrants have a **shorter credit history**
     - Length of credit report has positive impact to credit scores ($\approx$15% FICO)

Institutional Details On Lending In America

- **Information frictions** may reduce immigrant credit access:
  1. Immigrants have **no credit report on arrival**
     - **Lender's prior:** Consumer with no U.S. credit report is **high credit risk**
  2. After entering the credit system, immigrants have a **shorter credit history**
     - Length of credit report has positive impact to credit scores ($\approx$15% FICO)

- Different cultural **tastes** & informational barriers for consumers
  - Differences in familiarity & willingness to use debt, understanding of how to build credit score

- Lenders use credit scores ($+$ other information) in decisions
  - Can use immigration status, can't use protected characteristics (e.g., national origin, race) under Equal Credit Opportunity Act (ECOA)

Credit Market Entry

Main Cross-Sectional Regression Specifications

β_1: average difference for Immigrants vs. Non-Immigrants from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \tag{1}$$

- Includes fixed effects: birth year ($\mu_{b(i)}$) and geography based on first ZIP5 ($\eta_{g(i)}$)

Main Cross-Sectional Regression Specifications

β_1: average difference for Immigrants vs. Non-Immigrants from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \tag{1}$$

- Includes fixed effects: birth year ($\mu_{b(i)}$) and geography based on first ZIP5 ($\eta_{g(i)}$)

β_2: marginal difference for SSN Age one year later, from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \tag{2}$$

- Cluster standard errors by birth year × ZIP2

Main Cross-Sectional Regression Specifications

β_1: average difference for Immigrants vs. Non-Immigrants from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \tag{1}$$

- Includes fixed effects: birth year ($\mu_{b(i)}$) and geography based on first ZIP5 ($\eta_{g(i)}$)

β_2: marginal difference for SSN Age one year later, from OLS regression:

$$Y_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \beta_2 \cdot SSN\ Age_i + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i \tag{2}$$

- Cluster standard errors by birth year $\times$ ZIP2

Results robust to including additional F.E.:

- Last ZIP5, Longest-held ZIP5, Number of ZIP5 (geographic mobility proxy), & credit score.

Also robust to restricting to consumers who have not emigrated by 2024.

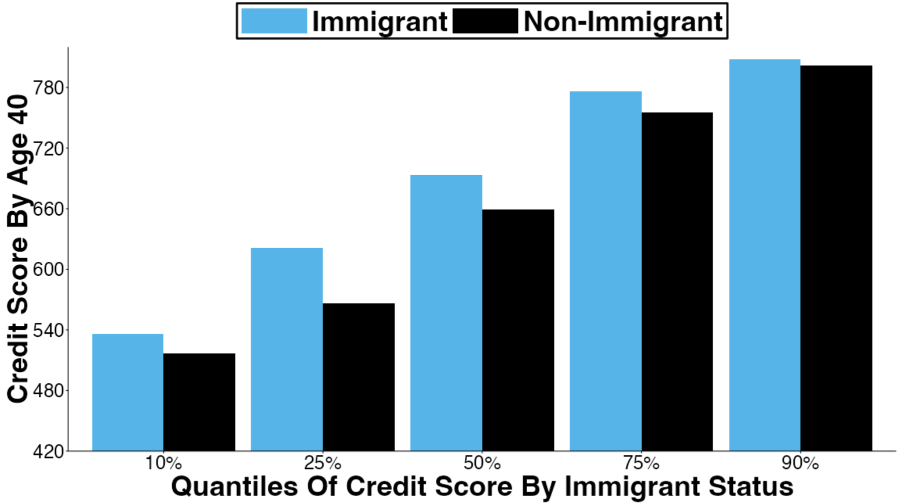
Immigrants, & Older SSN Age Immigrants, Enter Credit Market Later Than Non-Immigrants & Younger SSN Age Immigrants

	Age at First Credit Report	
	(1)	(2)
Immigrant	5.53***	2.29***
	(0.02)	(0.02)
SSN Age		0.75***
		(0.01)
F.E. Birth Year	X	X
F.E. First Zip5	X	X
R^2	0.193	0.209
Mean, SSN Age <21	20.19	20.19

$N = 6,297,340$, *** 0.5%, ** 1%, * 5% statistical significance

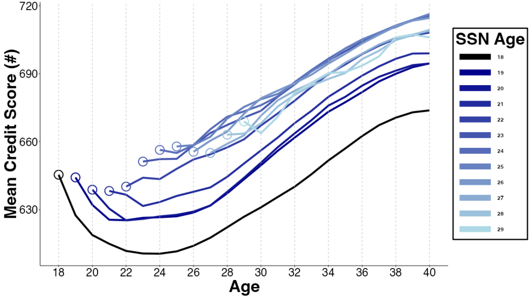
Creditworthiness

Once Scored, Immigrants (vs. Non-Immigrants) +26.7 Credit Score By Age 30



Credit Score: +17.2 points immigrant (vs. non-immigrant), ↑ 2.3 for each SSN Age

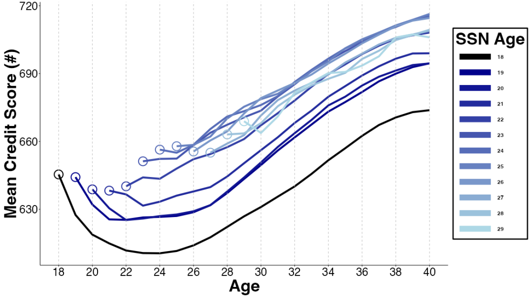
A. Mean Credit Score (Once Scored)



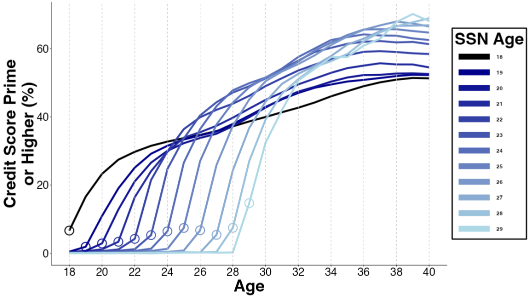
Once Scored, Immigrants (vs. Non-Immigrants) +26.7 Credit Score By Age 30

Credit Score: +17.2 points immigrant (vs. non-immigrant), ↑ 2.3 for each SSN Age

A. Mean Credit Score (Once Scored)



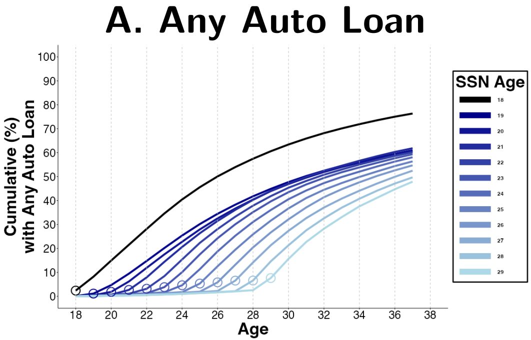
B. Prime or Higher Credit Score (660+)



↓ 5.5% delinquency rates conditional on credit score

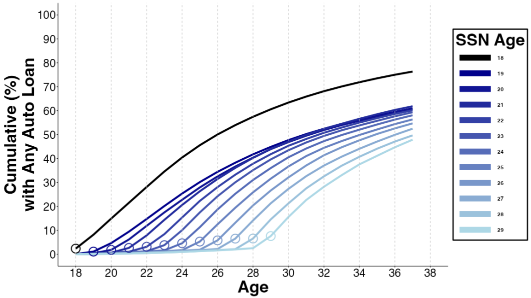
Credit Access

Immigrants Persistently Less Likely To Hold Auto Loans & Mortgages

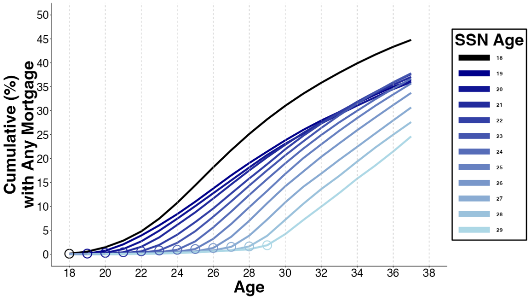


Immigrants Persistently Less Likely To Hold Auto Loans & Mortgages

A. Any Auto Loan



B. Any Mortgage



By Age 37, Immigrants 17% (16%) Less Likely To Have Auto Loan (Mortgage) - ↓ Auto Loan & Mortgage Use For Later In Life Immigrants

Dep. Var: By Age 37, has ...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-15.66^{***}	-8.69^{***}	-8.91^{***}	-1.05^{***}	-1.35	-0.37^{***}
	(0.18)	(0.23)	(0.17)	(0.23)	(0.08)	(0.11)
SSN Age		-1.62^{***}		-1.83^{***}		-0.23
		(0.04)		(0.04)		(0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.041	0.042	0.063	0.063	0.024	0.024
Mean, SSN Age <21	76.06	77.06	44.63	44.63	94.12	94.12

$N = 6,297,340$, *** 0.5%, ** 1%, * 5% statistical significance

“Paired Cohorts”: Comparing Immigrants vs. One Year Younger SSN Age (Immigrants With Same Birth Year)

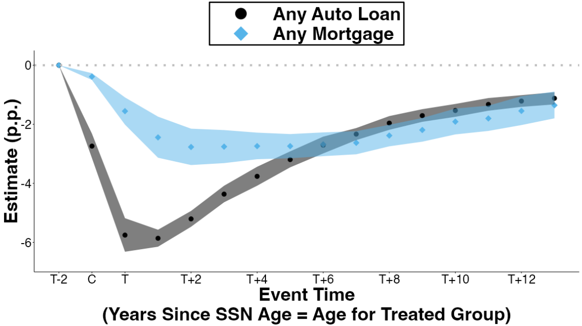
$$Y_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_\tau \left(\mathbb{I}\{focal\ cohort\}_{k(p)} \times \mathbb{I}\{t = \tau\} \right) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \quad (3)$$

- ‘Focal Cohort’: $SSN\ Age_{k'}$
- ‘Control Cohort’: $SSN\ Age_{k'} - 1$, same birth year as focal. 68 pairs (p)
- 2,176 cohort-by-year observations ($k(p), t$), s.e. wild cluster bootstrap

"Paired Cohorts": Comparing Immigrants vs. One Year Younger SSN Age (Immigrants With Same Birth Year)

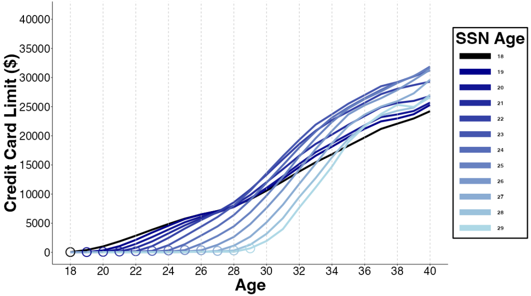
$$Y_{k(p),t} = \sum_{\tau=-1}^{+13} \delta_\tau \left(\mathbb{I}\{focal\ cohort\}_{k(p)} \times \mathbb{I}\{t=\tau\}\right) + \pi_{k(p)} + \nu_{p,t} + \epsilon_{k(p),t} \quad (3)$$

- 'Focal Cohort': $SSN\ Age_{k'}$
- 'Control Cohort': $SSN\ Age_{k'} - 1$, same birth year as focal. 68 pairs (p)
- 2,176 cohort-by-year observations $(k(p), t)$, s.e. wild cluster bootstrap
- **Results, 13 years post-immigration:**
  $-$1.12 p.p. ever had an auto loan
  $-$1.36 p.p. ever had a mortgage



Immigrants’ Credit Card Limits Are Lower Than Non-Immigrants

A. Credit Card Limits ($)

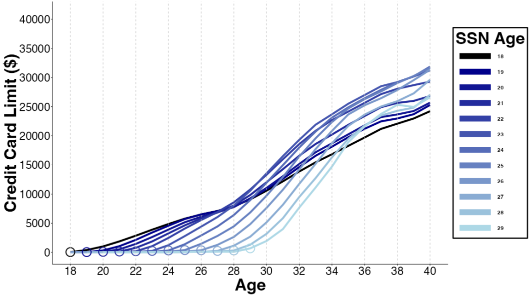


Similar patterns for number of credit cards,

credit card limit per card

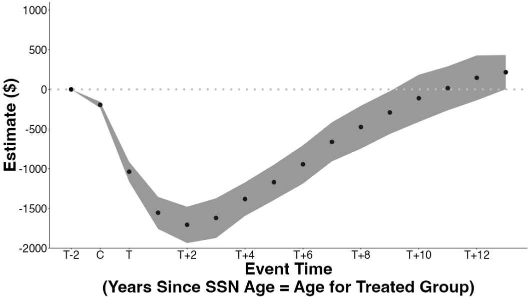
Immigrants' Credit Card Limits Are Lower Than Non-Immigrants

A. Credit Card Limits ($)



Similar patterns for number of credit cards, credit card limit per card

B. Paired Cohorts



8 Years Post-Immigration:

−$474 [95% C.I.: −$746, −$207]

Mechanisms

What Mechanisms Explain?
- Less Credit For More Creditworthy Immigrants
- Less Credit For Later In Life Immigrants

~~Creditworthiness~~

~~Emigration (Return Migration)~~

Credit Demand & Tastes

History Dependence

- Credit History Dependence: Financial friction...
  - no U.S. credit report
  - shorter credit history
- Non-Credit History Dependence: less time in U.S. means...
  - fewer years of U.S. earnings
  - less far along American life cycle

Immigrants Are 19% More Averse To Purchasing Cars With Debt (SCF, 2022)

Dep Var:	Auto Debt Attitude		Any Auto Debt		Any Auto Debt	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-15.41***	-23.02***	-4.27	-0.21	-11.14	-5.10
	(4.58)	(7.25)	(5.08)	(5.01)	(8.27)	(7.87)
Immigration Age		0.98			0.88	0.63
		(0.62)			(0.76)	(0.69)
Auto Debt Attitude				26.34***		26.24***
				(3.23)		(3.24)
Demographic Controls	X	X	X	X	X	X
R^2	0.049	0.050	0.063	0.106	0.063	0.106
Mean, Immigration Age <21	79.58	79.58	47.14	47.14	47.14	47.14

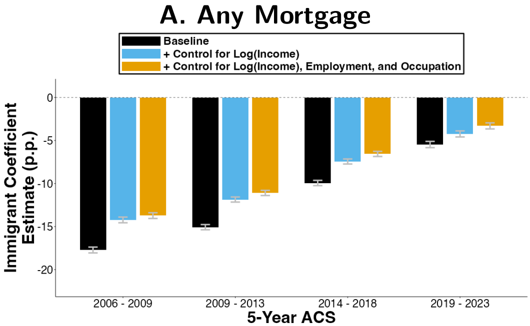
$N = 1,813$, *** 0.5%, ** 1%, * 5%, $^{+}$ 10% statistical significance

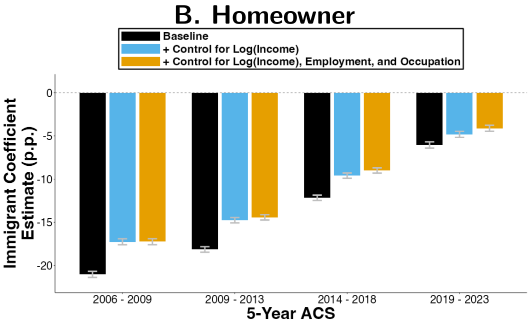
No clear evidence of more general negative attitude to debt

Immigrants & Later-in-Life Immigrants Have Lower Income (5-year ACS waves)

	Log (Income)							
	2006 - 2009		2009 - 2013		2014 - 2018		2019 - 2023	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-17.66***	-11.79***	-17.41***	-10.18***	-14.51***	-8.55***	-8.08***	-3.03***
	(0.36)	(0.67)	(0.31)	(0.56)	(0.28)	(0.53)	(0.34)	(0.64)
Immigration Age		-1.49***		-1.69***		-1.27***		-1.06***
		(0.14)		(0.11)		(0.10)		(0.12)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. Microdata Area	X	X	X	X	X	X	X	X
F.E. Sex	X	X	X	X	X	X	X	X
F.E. Education	X	X	X	X	X	X	X	X
F.E. Race	X	X	X	X	X	X	X	X
F.E. Hispanic	X	X	X	X	X	X	X	X
F.E. Survey Year	X	X	X	X	X	X	X	X
N	2,079,500	2,079,500	2,162,012	2,162,012	2,231,275	2,231,275	2,200,018	2,200,018
R^2	0.161	0.161	0.196	0.196	0.239	0.239	0.228	0.228
Mean Income, Immigration Age <21	$71,088	$71,088	$78,339	$78,339	$98,660	$98,660	$134,099	$134,099

Differences in Mortgage & Homeownership Remain After Controlling for Income & Demographics (5-year ACS waves)





Immigrating one year later in life leads to persistently lower mortgage use & homeownership

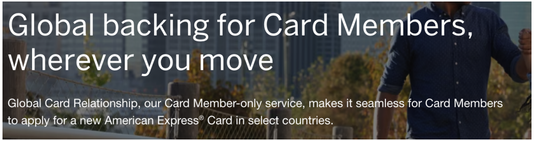
Average immigrant vs. non-immigrant difference in car ownership disappears over time

All regressions include fixed effects for...

birth year, geographic area, sex, education levels, race, hispanic, & survey year

Non-Portability Of Credit History Across National Boundaries May Prevent Profitable Lending. Recent Market Solutions?

American Express Global Card Relationship



Nova Credit

International credit data

Credit Passport® gives you access to foreign consumer-permissioned credit data that's translated to a local equivalent score to serve the 2 million immigrants that arrive in the United States each year and 280 million immigrants worldwide.

Equifax Canada Global Consumer Credit File

Global Consumer Credit File

Helping you grow and retain your clients who are newcomers to Canada

Global Consumer Credit File provides lenders with a seamless and secure platform that connects with global credit bureaus and enables accurate and informed decisioning through robust data.

Conclusions

How Do Immigrants Assimilate Into The Formal U.S. Consumer Credit System?

Study **selection** & **assimilation dynamics** over life cycle of Immigrants vs. Non-Immigrants
- exploit variation in age of immigration in credit data matched with immigration information

Immigrants vs. same age Non-Immigrants & Immigrants arriving at older ages:

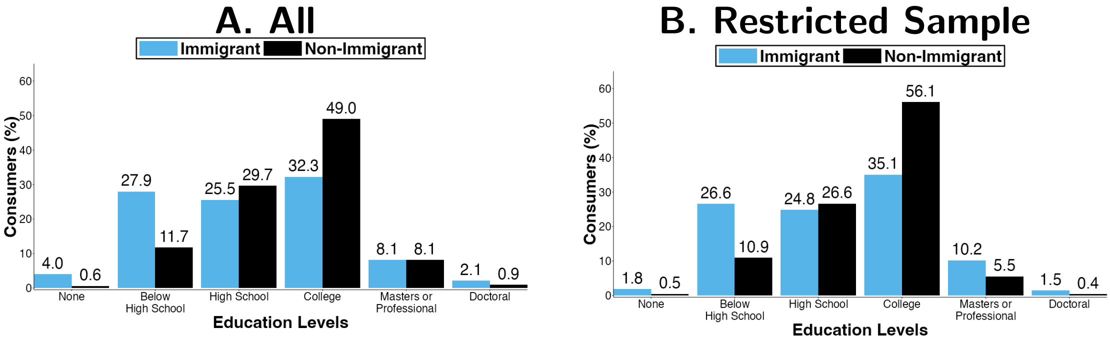
- **Positively selected**: more creditworthy (once scored)
- Despite better creditworthiness, **lower long-term access** to auto loans & mortgages
  - Lower mortgage use is largely explained by immigration timing
  - Lower auto loan use also due to auto loan-specific debt aversion
  - Not explained by emigration or income
- *Timing* of immigration best explains most of our results ("**history dependence**")

Thank You!

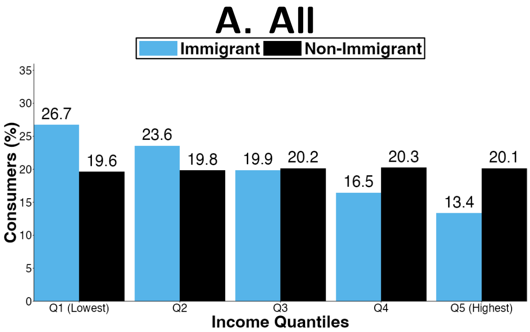
www.benedictgk.com

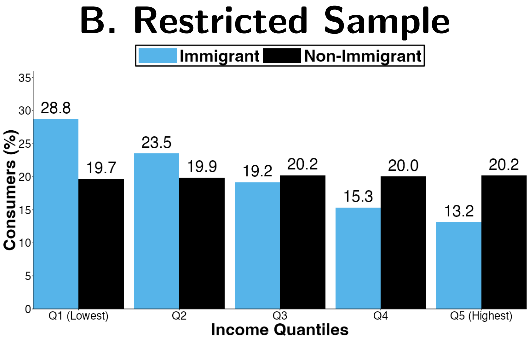
benedictgk@rice.edu

Distribution of Education in 5-year ACS (2010)



Distribution of Income in 5-year ACS (2010)





Sample In BTCCP

Sample Restrictions

- Birth years 1975 - 1987
- SSN Ages 0 to 29 (21 to 29 "Immigrants")

Entrant Sample

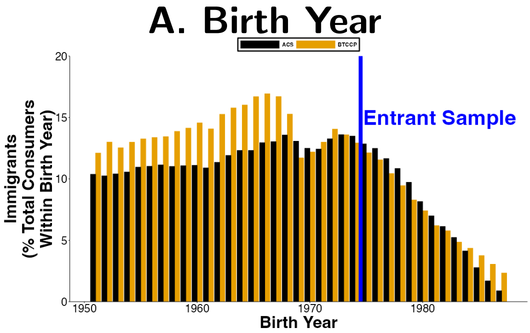
6,299,287 consumers

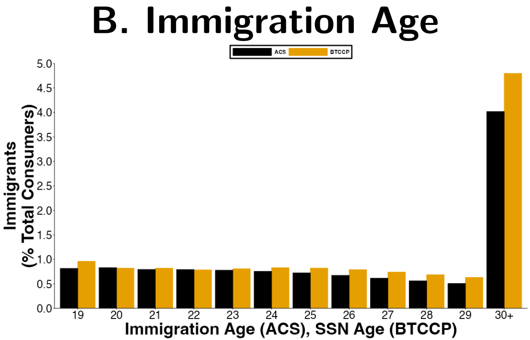
– 384,396 (6.1%) "Immigrants"

Follow **credit outcomes** in BTCCP for **ages 18 to 40**

Immigrant Classification Aligns With American Community Survey (ACS)

Distribution of Immigrants by...





Distribution of immigrants aligns by geography.

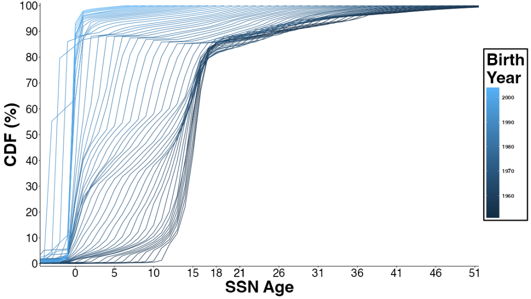
Immigrants also more geographically mobile than non-immigrants.

“Entrant Sample”: Restrict To Birth Years 1975 - 1987 To Observe U.S. Entry At Ages 21 To 29 & Credit Outcomes For Ages 18 To 40

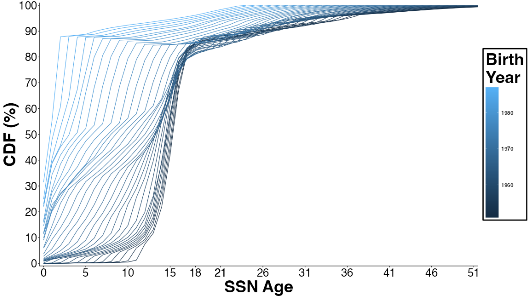
TransUnion Consumers	
... born 1951 to 2004	31,869,445
... with a SSN in TransUnion	25,237,917
Clean Sample	19,019,191
... SSN Age <21	16,793,939
... SSN Age 21+	2,225,252 (11.70%)
... Immigrant Cohorts (Birth Year x SSN Year)	775
Entrant Sample (SSN Age < 30)	6,299,287
... SSN Age < 21	5,914,891
... SSN Age 21-29	384,396 (6.10%)
... Immigrant Cohorts (Birth Year x SSN Year)	102

SSN Ages By Birth Year

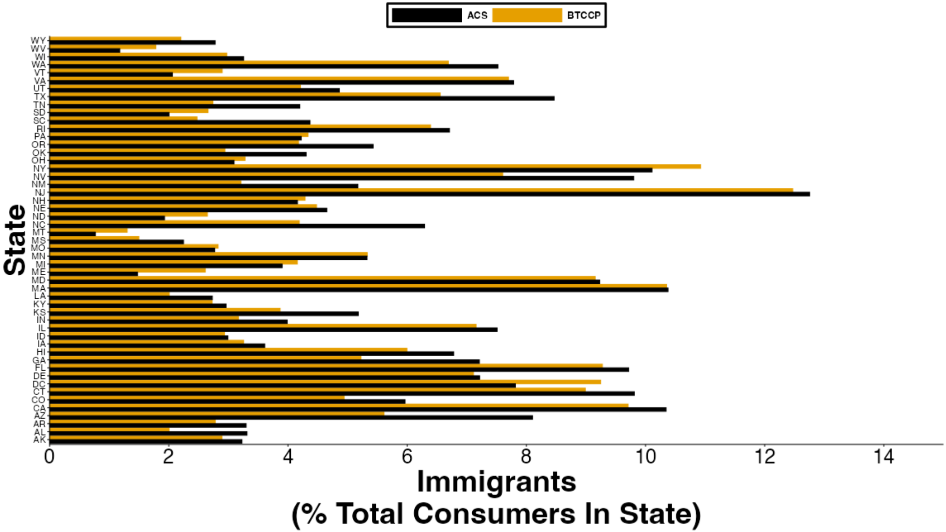
A. Before Cleaning



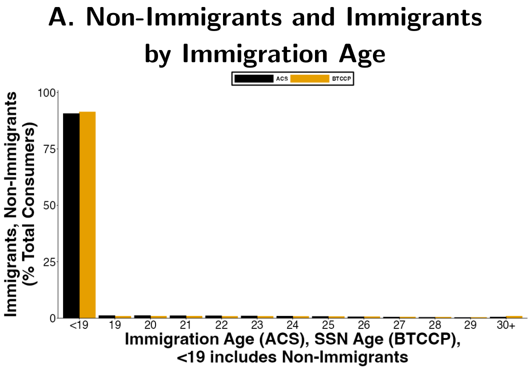
B. After Cleaning

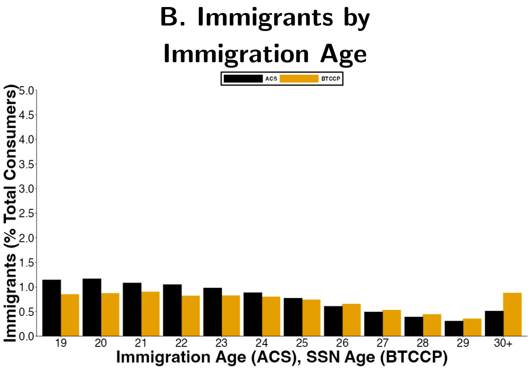


Geographic Comparison of Immigrants: ACS vs. BTCCP

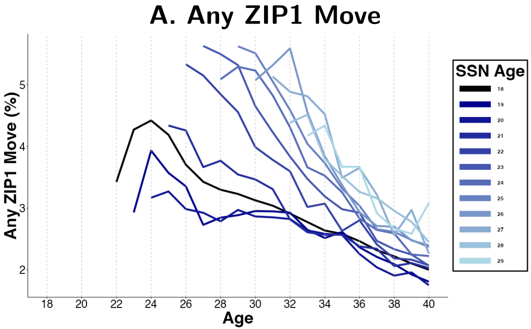


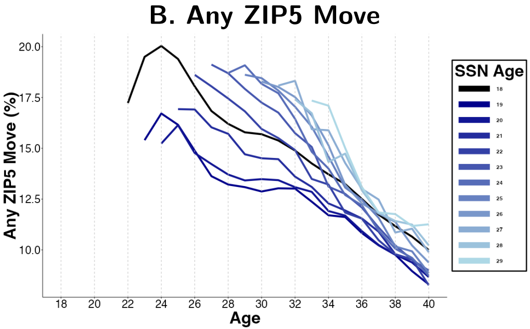
Immigrant Classification in Entrants Sample: ACS vs. BTCCP





Geographic Mobility





Consumers By SSN Age

SSN Age	Clean Sample	Entrant Sample
$<$19	16,455,525	5,805,684
19	182,481	53,950
20	155,933	55,257
21	155,743	57,114
22	149,210	51,894
23	153,427	52,314
24	157,685	50,705
25	156,152	46,924
26	150,045	41,481
27	140,610	33,525
28	130,243	28,040
29	119,788	22,399
30+	912,349	55,704

Sample Frame and Timing of SSN Assignment

Calendar year	Birth year 1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987
1993	≤18												
1994	19	≤18											
1995	20	19	≤18										
1996	21	20	19	≤18									
1997	22	21	20	19	≤18								
1998	23	22	21	20	19	≤18							
1999	24	23	22	21	20	19	≤18						
2000	25	24	23	22	21	20	19	≤18					
2001	26	25	24	23	22	21	20	19	≤18				
2002	27	26	25	24	23	22	21	20	19	≤18			
2003	28	27	26	25	24	23	22	21	20	19	≤18		
2004	29	28	27	26	25	24	23	22	21	20	19	≤18	
2005	30	29	28	27	26	25	24	23	22	21	20	19	≤18
2006	31	30	29	28	27	26	25	24	23	22	21	20	19
2007	32	31	30	29	28	27	26	25	24	23	22	21	20
2008	33	32	31	30	29	28	27	26	25	24	23	22	21
2009	34	33	32	31	30	29	28	27	26	25	24	23	22
2010	35	34	33	32	31	30	29	28	27	26	25	24	23
2011	36	35	34	33	32	31	30	29	28	27	26	25	24
2012	37	36	35	34	33	32	31	30	29	28	27	26	25
2013	38	37	36	35	34	33	32	31	30	29	28	27	26
2014	39	38	37	36	35	34	33	32	31	30	29	28	27
2015	40	39	38	37	36	35	34	33	32	31	30	29	28
2016	41	40	39	38	37	36	35	34	33	32	31	30	29
2017	42	41	40	39	38	37	36	35	34	33	32	31	30
2018	43	42	41	40	39	38	37	36	35	34	33	32	31
2019	44	43	42	41	40	39	38	37	36	35	34	33	32
2020	45	44	43	42	41	40	39	38	37	36	35	34	33
2021	46	45	44	43	42	41	40	39	38	37	36	35	34
2022	47	46	45	44	43	42	41	40	39	38	37	36	35
2023	48	47	46	45	44	43	42	41	40	39	38	37	36
2024	49	48	47	46	45	44	43	42	41	40	39	38	37

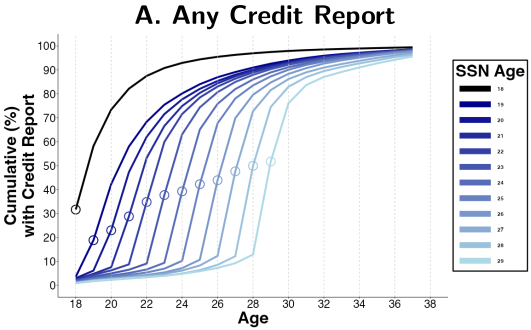
Age at SSN assignment

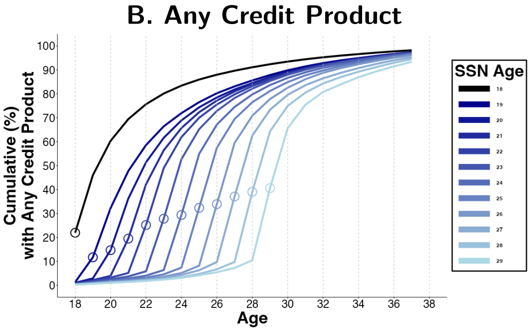
Years of data in our sample

Summary Statistics

	Non-Immigrants (*SSN Age < 21*)		Immigrants (*SSN Age 21 to 29*)	
	Mean	*N*	Mean	*N*
Credit Report				
Age at First Credit Report	20.19	5,914,891	26.09	384,396
Credit Score				
Any Score by Age 30 (%)	96.26	5,914,891	81.63	384,396
Credit Score by Age 30	624.2	5,693,584	658.3	313,799
Any Score by Age 40 (%)	99.35	5,914,891	97.94	384,396
Credit Score by Age 40	658.1	5,876,224	687.5	376,461
Auto Loan				
Any Auto Loan (%)	81.14	5,914,891	65.62	384,396
Age at First Auto Loan	26.13	4,799,057	30.59	252,243
Mortgage				
Any Mortgage (%)	50.28	5,914,891	44.42	384,396
Age at First Mortgage	29.53	2,973,846	32.88	170,754
Credit Card				
Any Credit Card (%)	96.87	5,914,891	97.39	384,396
Age at First Credit Card	22.33	5,730,022	27.20	374,365
Credit Card Limits at Age 30	$11,492	5,914,891	$10,211	384,396
Credit Card Limits at Age 40	$22,112	4,579,356	$26,608	342,258

Age At First Credit Report and First Credit Product By SSN Age Cohort

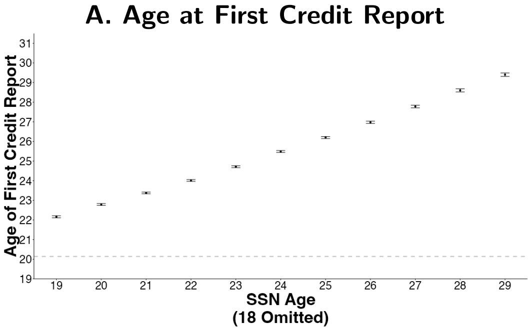


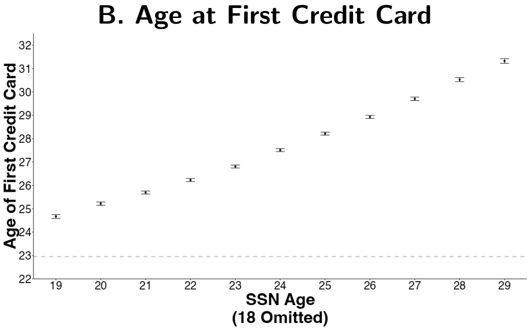


Credit Entry

Dep Var: Age at First...	Credit Report		Credit Product	
	(1)	(2)	(3)	(4)
Immigrant	5.53***	2.29***	5.20***	2.07***
	(0.02)	(0.02)	(0.02)	(0.03)
SSN Age		0.75***		0.73***
		(0.01)		(0.01)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.193	0.209	0.118	0.126
N	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	20.19	20.19	21.56	21.56

Age At First Credit Report & Credit Card

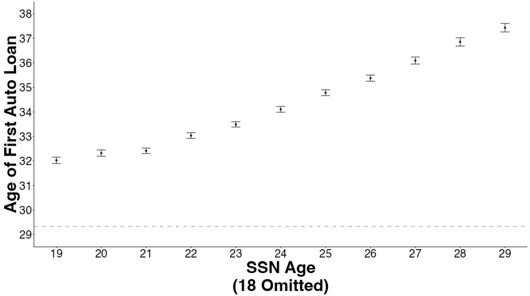




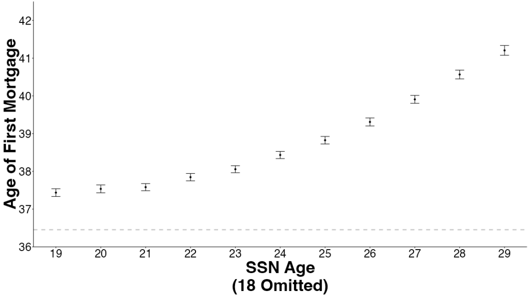
$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i$$

Age At First Auto Loan and Mortgage

A. Age at First Auto Loan



B. Age at First Mortgage

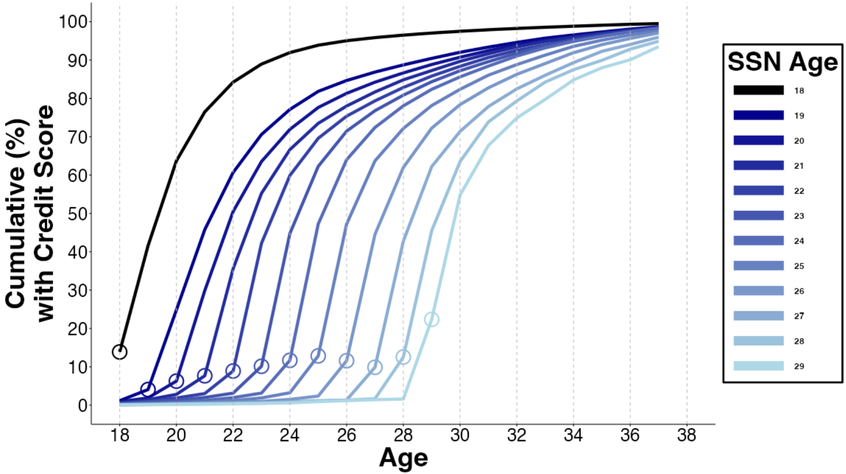


$$Y_i = \phi_{SSN\ Age(i)} + \mu_{b(i)} + \eta_{g(i)} + \epsilon_i$$

Credit Entry With Additional F.E.

Dep Var: Age at First...	Credit Report		Credit Product		Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	5.15***	2.07***	4.73***	1.81***	3.88***	1.60***	1.72***	0.03	4.21***	1.49***
	(0.02)	(0.02)	(0.02)	(0.03)	(0.04)	(0.04)	(0.03)	(0.04)	(0.02)	(0.03)
SSN Age		0.72***		0.68***		0.53***		0.39***		0.63***
		(0.01)		(0.01)		(0.01)		(0.01)		(0.01)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X	X	X
R^2	0.260	0.275	0.200	0.207	0.180	0.181	0.158	0.159	0.173	0.176
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	20.19	20.19	21.56	21.56	29.40	29.40	36.49	36.49	22.99	22.99

Lifecyle of Any Credit Score (Cumulative) By SSN Age Cohorts

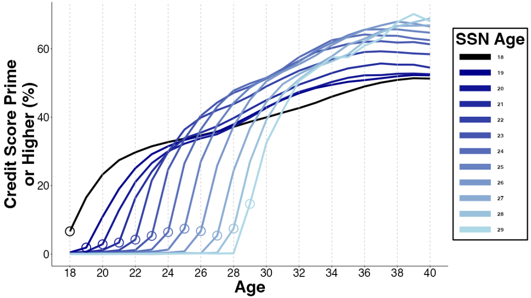


Any Credit Score At Ages 30 & 40

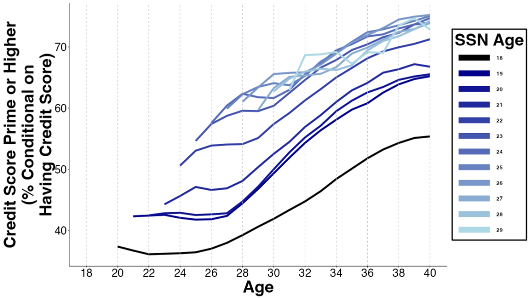
Dep Var: Any Credit Score by ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	-13.87***	-0.32	-1.23***	-0.6***
	(0.16)	(0.22)	(0.04)	(0.05)
SSN Age		-3.15***		-0.15***
		(0.06)		(0.01)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.045	0.053	0.009	0.009
N	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age $<$21	96.26	96.26	99.35	99.35

Prime of Higher (660+) Credit Score

A.

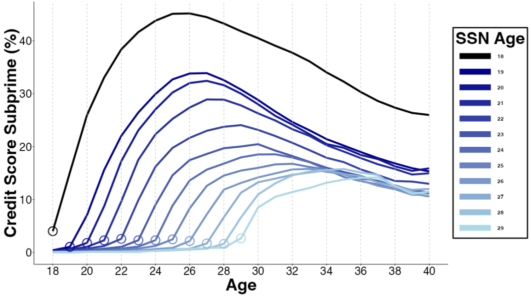


B. Conditional On Any Score)

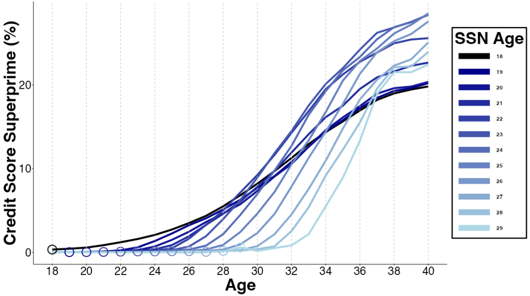


Lifecycle Of Any Subprime & Superprime Credit Scores By SSN Age

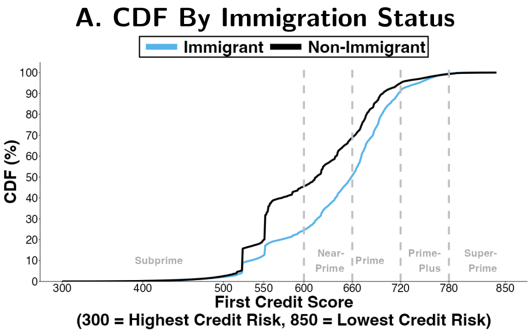
A. Subprime (<600)

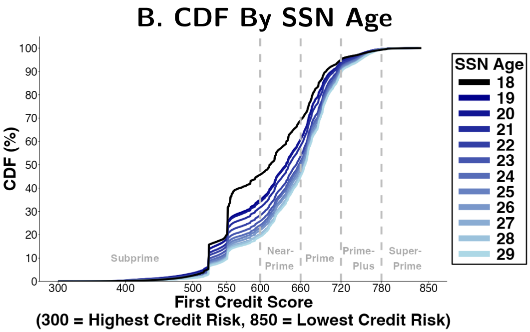


B. Superprime (780+)

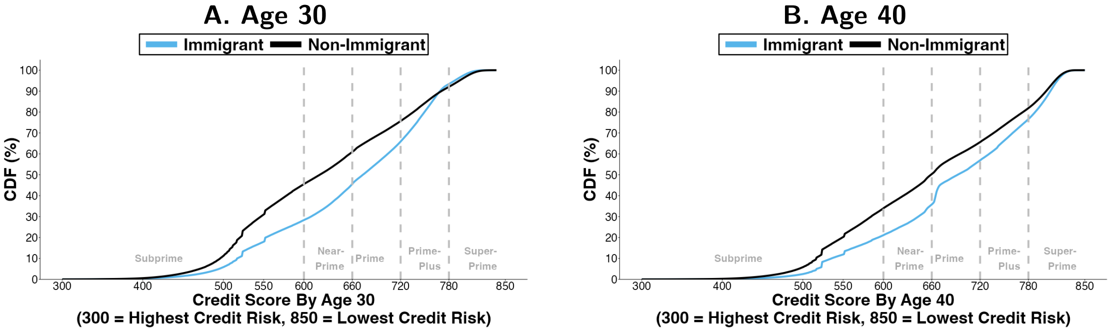


CDF Of First Credit Score By Immigration Status & SSN Age

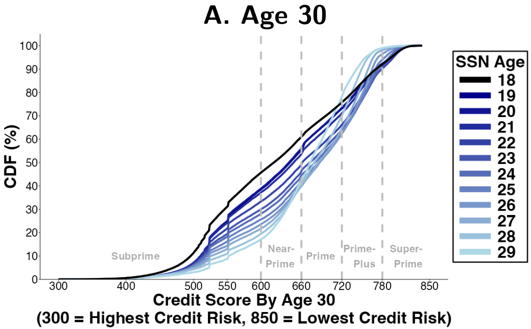


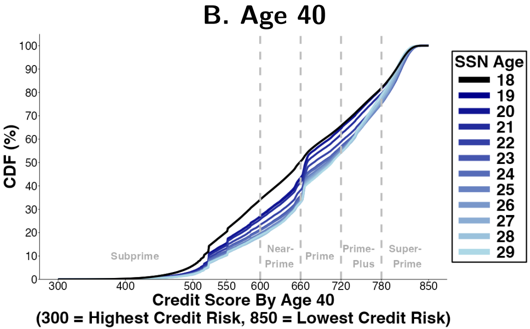


CDF Of Credit Score By Ages 30 & 40 By Immigration Status



CDF Of Credit Score By Ages 30 & 40 By SSN Age





Credit Scores At Ages 30 & 40

Dep Var: Credit Score by ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	26.7***	17.2***	24.6***	14.2***
	(0.4)	(0.6)	(0.4)	(0.5)
SSN Age		2.3***		2.4***
		(0.1)		(0.1)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.135	0.135	0.124	0.124
N	6,005,430	6,005,430	6,250,733	6,250,733
Mean, SSN Age $<$21	624.2	624.2	658.1	658.1

Credit Score Prime or Higher (660+) At Ages 30 & 40

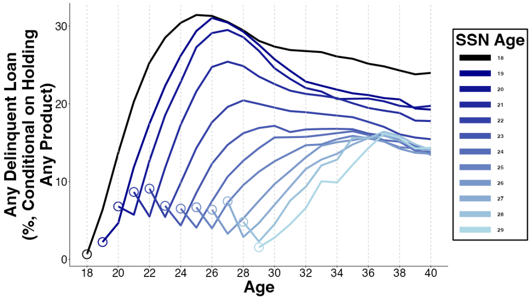
Dep Var: Prime or Higher by ...	Age 30		Age 40	
	(1)	(2)	(3)	(4)
Immigrant	4.61***	6.82***	11.24***	7.04***
	(0.17)	(0.25)	(0.17)	(0.22)
SSN Age		-0.51***		0.98***
		(0.05)		(0.04)
F.E. Birth Year	X	X	X	X
F.E. First Zip5	X	X	X	X
R^2	0.102	0.102	0.098	0.098
N	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age $<$21	37.63	37.63	49.33	49.33

Credit Scores At Ages 30 & 40, With Additional F.E.

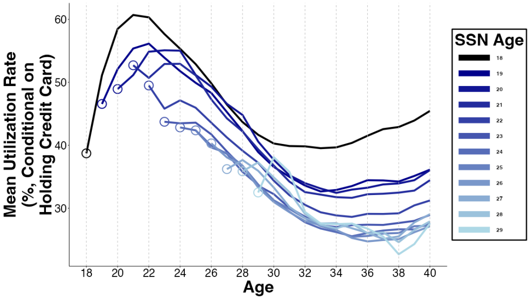
	Credit Score by...				Probability of Prime or Higher Score by...			
	Age 30		Age 40		Age 30		Age 40	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	22.21***	14.48***	22.99***	12.59***	3.78***	6.05***	10.52***	6.33***
	(0.36)	(0.54)	(0.34)	(0.46)	(0.15)	(0.22)	(0.14)	(0.20)
SSN Age		1.91***		2.43***		-0.53***		0.98***
		(0.10)		(0.09)		(0.05)		(0.04)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X
R^2	0.202	0.202	0.191	0.191	0.155	0.155	0.149	0.149
N	6,001,282	6,001,282	6,246,734	6,246,734	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	624.21	624.21	658.13	658.13	37.63	37.63	49.33	49.33

Immigrants (vs. Non-Immigrants) Have More Creditworthy Behaviors: Lower Delinquency Rates & Lower Credit Card Utilization

A. Any Delinquency

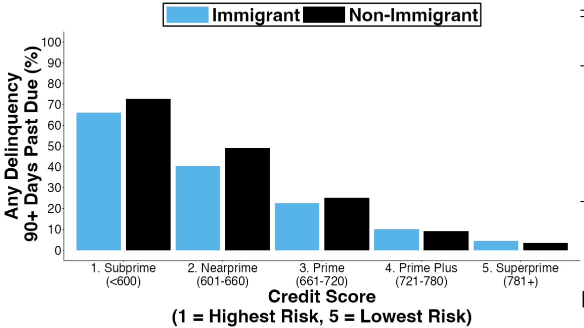


B. Mean Credit Card Utilization Rate



↓ 5% delinquency rates at age 37 conditional on credit score at age 30

Immigrants (vs. Non-Immigrants) Are More Creditworthy: ↓5% Delinquency Rates At Age 37 Conditional On Age 30 Credit Score



	(1)	(2)
Immigrant	-1.50^{***}	-1.23^{***}
	(0.25)	(0.39)
SSN Age	-0.07	
		(0.05)
F.E. Birth Year	X	X
F.E. Age 30 Zip5	X	X
F.E. Age 30 Credit Score	X	X
R^2	0.185	0.185
N	5,804,866	5,804,866
Mean, SSN Age <21	27.18	27.18

Calibration Bias: Delinquency At 37 Conditional On Age 30 Credit Score

Dep Var: Any	All		Subprime		Nearprime		Prime		Prime Plus		Superprime	
delinquency by age 37	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-1.50***	-1.23***	-2.64***	-3.31***	-3.51***	-2.65***	-1.44***	0.18	0.13	1.08***	0.29*	0.42
	(0.10)	(0.16)	(0.21)	(0.36)	(0.23)	(0.41)	(0.16)	(0.31)	(0.09)	(0.17)	(0.11)	(0.23)
SSN Age		-0.07*		0.18*		-0.20*		-0.36***		-0.22***		-0.04
		(0.03)		(0.08)		(0.08)		(0.06)		(0.03)		(0.06)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Zip5	X	X	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Credit Score	X	X	X	X	X	X	X	X	X	X	X	X
R^2	0.185	0.185	0.032	0.032	0.028	0.028	0.027	0.027	0.023	0.023	0.016	0.016
N	5,802,755	5,802,755	2,534,973	2,534,973	900,676	900,676	876,362	876,362	1,003,211	1,003,211	465,391	465,391
Mean, SSN Age <21	27.18	27.18	45.45	45.45	26.17	26.17	13.11	13.11	4.52	4.52	1.63	1.63

Credit Access

Dep Var: Age at First...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	4.96***	2.27***	2.26***	0.40*	4.80***	1.79***
	(0.05)	(0.05)	(0.04)	(0.04)	(0.02)	(0.03)
SSN Age		0.63***		0.43***		0.70***
		(0.01)		(0.01)		(0.01)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.083	0.084	0.082	0.083	0.082	0.086
N	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age $<$21	29.40	29.40	36.49	36.49	22.99	22.99

Credit Access With Additional F.E.

Dep Var: Age at First...	Credit Report		Credit Product		Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	5.15***	2.07***	4.73***	1.81***	3.88***	1.60***	1.72***	0.03	4.21***	1.49***
	(0.02)	(0.02)	(0.02)	(0.03)	(0.04)	(0.04)	(0.03)	(0.04)	(0.02)	(0.03)
SSN Age		0.72***		0.68***		0.53***		0.39***		0.63***
		(0.01)		(0.01)		(0.01)		(0.01)		(0.01)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X	X	X
R^2	0.260	0.275	0.200	0.207	0.180	0.181	0.158	0.159	0.173	0.176
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age <21	20.19	20.19	21.56	21.56	29.40	29.40	36.49	36.49	22.99	22.99

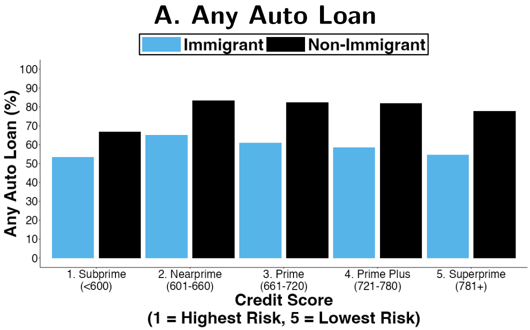
Credit Access By Age 37

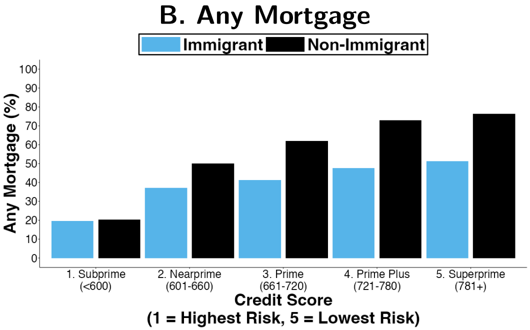
Dep. Var: By Age 37, has ...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-15.66***	-8.69***	-8.91***	-1.05***	-1.35***	-0.37***
	(0.18)	(0.23)	(0.17)	(0.23)	(0.08)	(0.11)
SSN Age		-1.62***		-1.83***		-0.23***
		(0.04)		(0.04)		(0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
R^2	0.041	0.042	0.063	0.063	0.024	0.024
N	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340	6,297,340
Mean, SSN Age <21	76.06	76.06	44.63	44.63	94.12	94.12

Credit Access By Age 37 With Additional F.E.

Dep Var: By Age 37, has...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-10.61***	-5.62***	-5.21***	1.39***	0.09	0.37***
	(0.17)	(0.21)	(0.17)	(0.21)	(0.07)	(0.10)
SSN Age		-1.16***		-1.54***		-0.06***
		(0.04)		(0.04)		(0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.137	0.137	0.151	0.151	0.070	0.070
N	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372	6,293,372
Mean, SSN Age $<$21	76.06	76.06	44.63	44.63	94.12	94.12

Any Auto Loan & Mortgage By Age 37 Conditional On Age 30 Credit Score

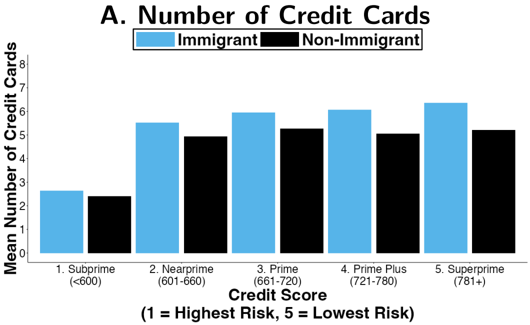


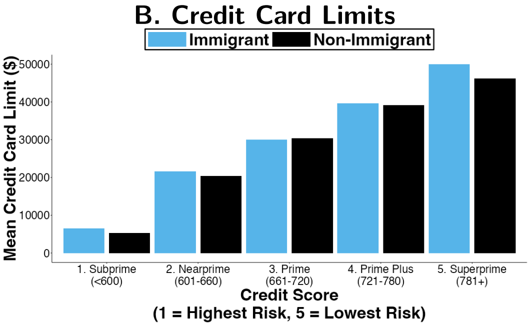


Age 37 Credit Outcomes Conditional On Age 30 Credit Score

Dep Var: Outcomes at Age 37...	Any Auto Loan		Any Mortgage		Any Credit Card		Number of Credit Cards		Credit Card Limits	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	-15.01***	-9.50***	-12.05***	-2.71***	-1.55***	-1.85***	0.36***	0.17***	-937.9***	732.3***
	(0.20)	(0.24)	(0.20)	(0.25)	(0.08)	(0.12)	(0.01)	(0.02)	(147.1)	(179.4)
SSN Age		-1.35***		-2.29***		0.07***		0.05***		-409.3***
		(0.04)		(0.05)		(0.02)		(0.00)		(35.1)
F.E. Birth Year	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Zip5	X	X	X	X	X	X	X	X	X	X
F.E. Age 30 Credit Score	X	X	X	X	X	X	X	X	X	X
R^2	0.108	0.108	0.267	0.268	0.110	0.110	0.141	0.141	0.305	0.305
N	5,802,755	5,802,755	5,802,755	5,802,755	5,802,755	5,802,755	5,802,755	5802755	5,802,755	5,802,755
Mean, SSN Age <21	75.1	75.1	44.54	44.54	92.06	92.06	3.9	3.9	$20,431.9	$20,431.9

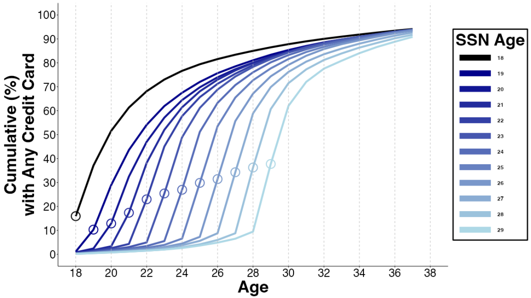
Credit Cards By Age 37 Conditional On Age 30 Credit Score



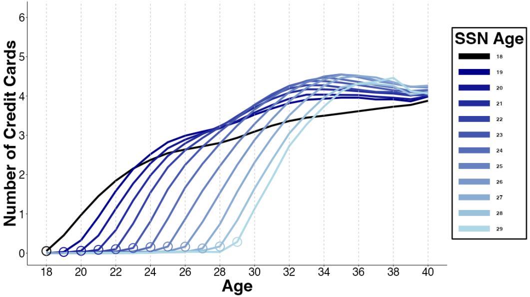


Immigrants' Access to Credit Cards Converges With Non-Immigrants

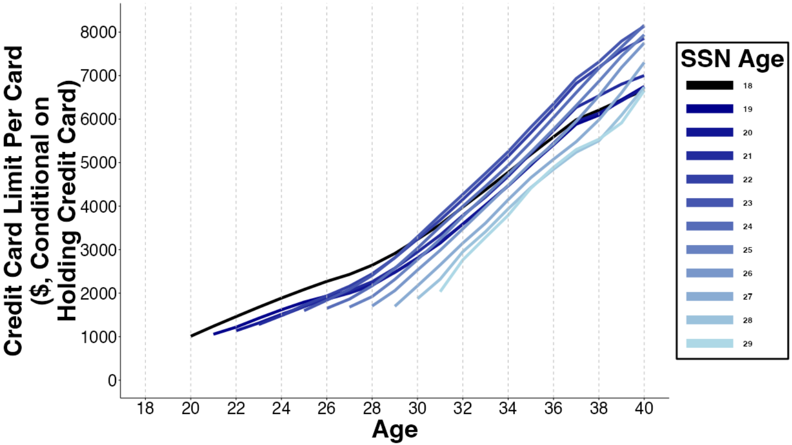
A. Any Credit Card



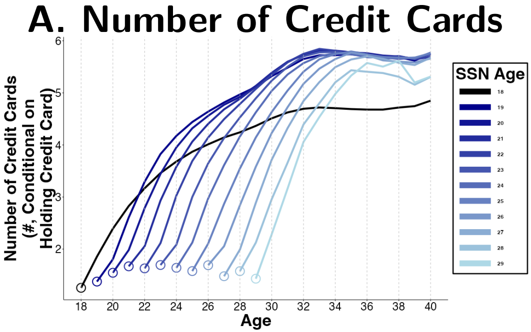
B. Number of Credit Cards

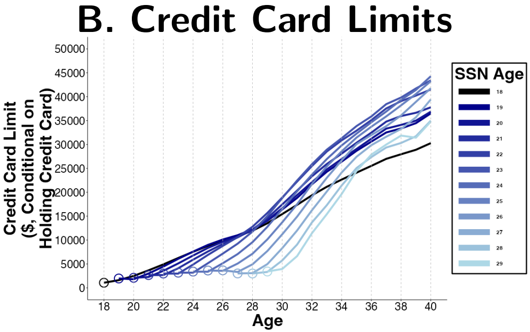


Credit Card Limit Per Card



Immigrants' Credit Card Access Conditional On Holding Any Credit Card

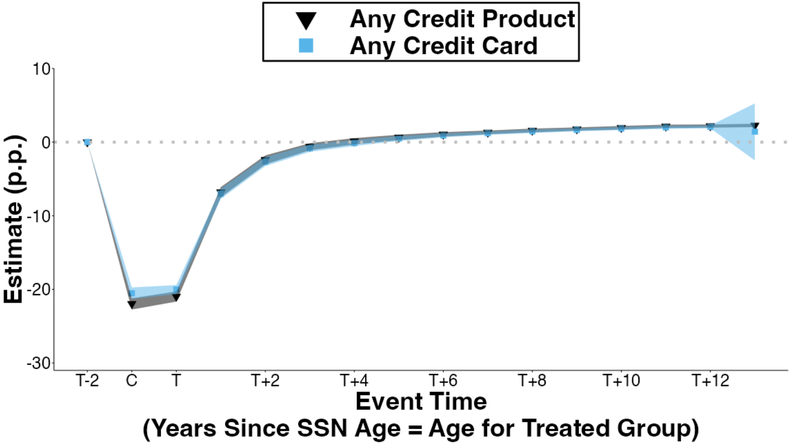




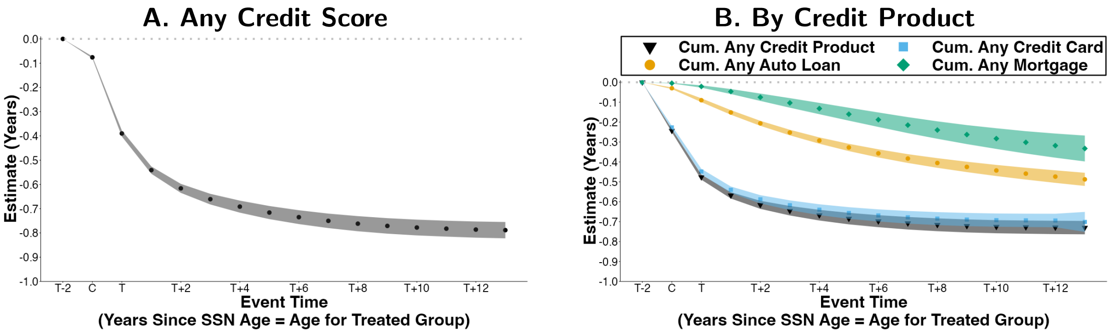
Credit Card Outcomes At Age 30 & At Age 40

	Number of Credit Cards At...				Credit Card Limits At...			
	Age 30		Age 40		Age 30		Age 40	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-0.13***	1.12***	0.57***	0.15***	-2,748.6***	3,027.2***	2,296.3***	1,539.3***
	(0.02)	(0.02)	(0.02)	(0.02)	(98.0)	(90.8)	(140.5)	(167.5)
SSN Age		-0.29***		0.09***		-1,345.2***		168.5***
		(0.00)		(0.00)		(21.0)		(32.0)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X	X	X
R^2	0.126	0.128	0.096	0.096	0.141	0.143	0.182	0.182
N	6,293,372	6,293,372	4,914,884	4,914,884	6,293,372	6,293,372	4,914,884	4,914,884
Mean, SSN Age $<$21	3.35	3.35	3.92	3.92	\$11,491.7	\$11,491.7	\$22,112.0	\$22,112.0

Paired Cohorts: First Credit Product & First Credit Card



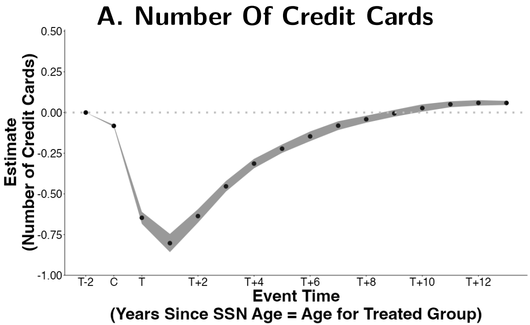
Paired Cohorts: Cumulatives

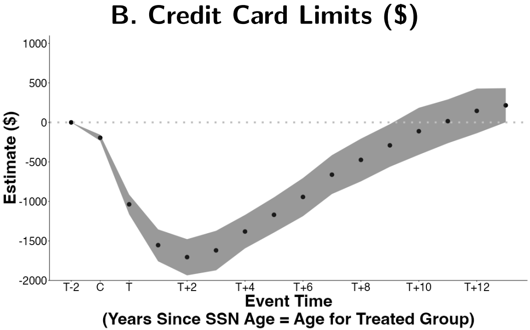


Paired Cohorts: Cumulative Years of Delay

Dep. Var: Any ...	Credit Score (1)	Credit Product (2)	Auto Loan (3)	Mortgage (4)	Credit Card (5)
C	-0.076***	-0.243***	-0.03***	-0.005***	-0.227***
	[-0.080, -0.072]	[-0.251, -0.235]	[-0.034, -0.026]	[-0.006, -0.003]	[-0.235, -0.218]
T	-0.390***	-0.476***	-0.09***	-0.021***	-0.448***
	[-0.403, -0.377]	[-0.490, -0.463]	[-0.100, -0.080]	[-0.027, -0.015]	[-0.463, -0.434]
$T+5$	-0.716***	-0.684***	-0.327***	-0.160***	-0.656***
	[-0.745, -0.686]	[-0.717, -0.653]	[-0.347, -0.308]	[-0.195, -0.126]	[-0.689, -0.625]
$T+10$	-0.778***	-0.725***	-0.443***	-0.283***	-0.692***
	[-0.813, -0.741]	[-0.762, -0.689]	[-0.473, -0.412]	[-0.338, -0.228]	[-0.729, -0.656]
N Paired Cohorts	68	68	68	68	68
N Consumers	463,542	463,542	463,542	463,542	463,542
N	7,416,672	7,416,672	7,416,672	7,416,672	7,416,672

Paired Cohorts: Credit Cards





Potential Mechanism: Emigration (Return Migration)

Proxy emigration by **attrition** from data

Immigrants exit data more than Non-Immigrants

- If arrive earlier, on average, exit data earlier, so can't explain paired cohort results

Restrict to consumers observed in 2024:

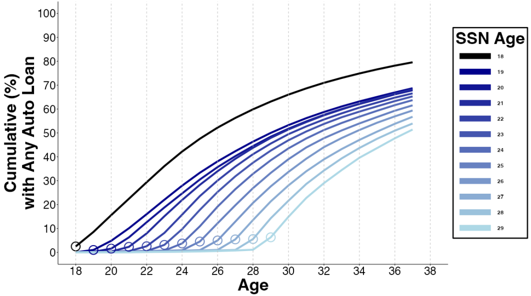
- **Mortgage** Immigrant vs. Non-Immigrant gap closes butSSN Age gap remains
- **Auto loan** Immigrant vs. Non-Immigrant gap & SSN Age gap remains
- **Credit card** dynamics by SSN Age remains

Consumers Observed In Data In 2024

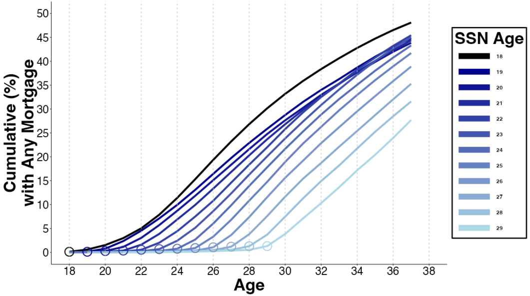
Dep Var: In 2024, has...	Any Credit Report		Any Credit Score		Any Credit Tradeline	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-2.10***	-4.38***	-6.25***	-11.19***	-7.39***	-11.83***
	(0.10)	(0.16)	(0.18)	(0.24)	(0.21)	(0.25)
SSN Age		0.53***		1.15***		1.04***
		(0.02)		(0.04)		(0.04)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.045	0.045	0.112	0.113	0.147	0.147
N	6,293,372	6,293,372	6293372	6,293,372	6,293,372	6,293,372
Mean, SSN Age $<$21	95.92	95.92	89.94	89.94	84.34	84.34

Lifecycle Of Credit For Consumers Observed In 2024

A. Any Auto Loan



B. Any Mortgage



Credit Access By Age 37 With Additional F.E. Conditional On Any Credit Score Observed in 2024

Dep Var: By Age 37, has...	Auto Loan		Mortgage		Credit Card	
	(1)	(2)	(3)	(4)	(5)	(6)
Immigrant	-10.75***	-3.98***	-4.48***	4.88***	-0.49***	0.14
	(0.17)	(0.22)	(0.18)	(0.23)	(0.07)	(0.09)
SSN Age		-1.56***		-2.15***		-0.14***
		(0.04)		(0.05)		(0.02)
F.E. Birth Year	X	X	X	X	X	X
F.E. First Zip5	X	X	X	X	X	X
F.E. Last Zip5	X	X	X	X	X	X
F.E. Longest Zip5	X	X	X	X	X	X
F.E. Number Zip5	X	X	X	X	X	X
R^2	0.11	0.111	0.138	0.138	0.061	0.061
N	5,624,238	5,624,238	5,624,238	5,624,238	5,624,238	5,624,238
Mean, SSN Age <21	79.38	79.38	48.03	48.03	95.38	95.38

Use Survey of Consumer Finances (SCF) 2022 To Evaluate Mechanisms

SCF 2022 includes if Immigrant & Age of Immigration

Adapt earlier cross-sectional regression to this data:

$$\mathrm{Y}_i = \beta_1 \cdot \mathbb{I}\{Immigrant\}_i + \beta_2 \cdot Immigration\ Age_i + \mu_{b(i)} + \eta_{d(i)} + \epsilon_i$$

$\eta_{d(i)}$ includes control for Log(Income) and F.E.: male, race, hispanic, spouse, marital status, household size, number of children, & education levels

SCF Debt Attitude Questions:

*"People have many different reasons for borrowing money which they pay back over a period of time. For each of the reasons I read, please tell me whether you feel it is all right for someone like yourself to borrow money...***to finance the purchase of a car***."*

Debt Attitudes (SCF, 2022)

Dep Var:	Auto Debt Attitude		Any Auto Debt		Any Auto Debt		General Debt Attitude	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-15.41***	-23.02***	-4.27	-11.14	-0.21	-5.10	-8.96	-15.94
	(4.58)	(7.25)	(5.08)	(8.27)	(5.01)	(7.87)	(7.92)	(13.71)
Immigration Age		0.98		0.88		0.63		0.89
		(0.62)		(0.76)		(0.69)		(1.19)
Auto Debt Attitude					26.34***	26.24***		
					(3.23)	(3.24)		
Demographic Controls	X	X	X	X	X	X	X	X
R^2	0.049	0.050	0.063	0.063	0.106	0.106	0.026	0.026
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age <21	79.58	79.58	47.14	47.14	47.14	47.14	3.80	3.80

*** 0.5%, ** 1%, * 5%, + 10% statistical significance

More Granular Debt Attitudes (SCF, 2022)

Dep Var: All Right To Borrow...	In General (Positive)		In General (Negative)		For Living Expenses		For Vacation		For Education	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	-4.27	-4.29	4.69	11.65	-8.76^{+}	-8.35	-5.01	-8.37	-2.80	9.07
	(4.83)	(8.95)	(4.65)	(7.09)	(4.89)	(7.66)	(3.40)	(5.21)	(4.18)	(5.59)
Immigration Age		0.00		-0.89		-0.05		0.43		-1.52*
		(0.80)		(0.62)		(0.74)		(0.56)		(0.67)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.016	0.016	0.043	0.044	0.032	0.032	0.020	0.020	0.040	0.045
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1813
Mean, Immigration Age <21	29.51	29.51	25.71	25.71	73.38	73.38	15.30	15.30	84.24	84.24

*** 0.5%, ** 1%, * 5%, $^{+}$ 10% statistical significance

Immigrants Have ↓ Demand for Debt (↑ Auto, ↑ Credit Cards, ↑ Mortgage. All Attenuated By Immigration Age) To Non-Immigrants (SCF, 2022) ...But Low Statistical Power To Detect Out Differences

Dep Var: Apply For...	Any Credit		Auto		Mortgage		Credit Card		Other Credit	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Immigrant	2.32	-1.08	2.03	-6.55	2.11	5.65	1.63	2.02	1.84	-7.66^{+}
	(4.56)	(7.17)	(4.37)	(7.53)	(3.8)	(6.85)	(3.22)	(5.32)	(3.62)	(4.61)
Immigration Age		0.44		1.10		-0.45		-0.05		1.22*
		(0.62)		(0.72)		(0.65)		(0.48)		(0.55)
Demographic Controls	X	X	X	X	X	X	X	X	X	X
R^2	0.051	0.051	0.051	0.053	0.045	0.045	0.045	0.045	0.026	0.029
N	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813	1,813
Mean, Immigration Age <21	64.81	64.81	19.95	19.95	11.28	11.28	9.28	9.28	12.08	12.08

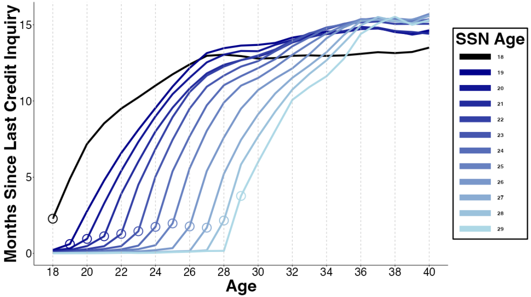
*** 0.5%, ** 1%, * 5%, $^{+}$ 10% statistical significance

Credit inquiries data in credit panel shows
↑ immigrant credit demand in 20s, ↓ demand in 30s

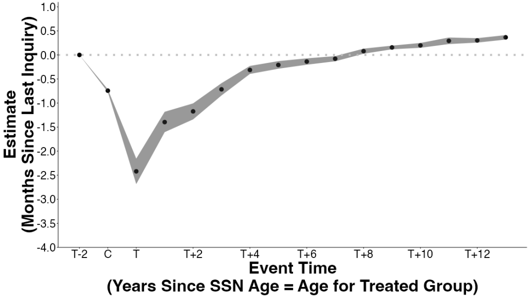
Does Lower Credit Demand Explain Lower Access?

Immigrants Have ↑ Credit Demand In 20s, ↓ Demand In 30s

A. Months Since Last Credit Inquiry
Higher Value = Lower Credit Demand



B. Paired Cohorts
Months Since Last Credit Inquiry



Caveat: Lower credit demand in 30s may be function of earlier rejections → deterrence

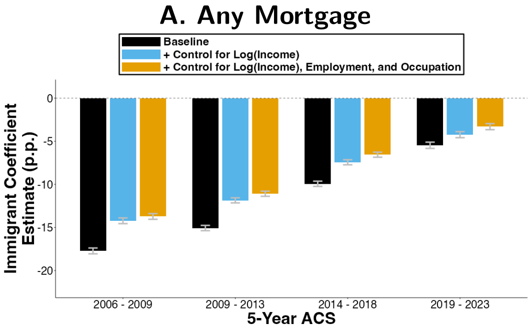
Income, Controlling for Demographics (ACS)

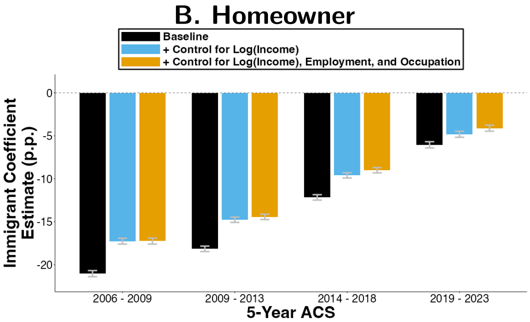
	Log (Income)							
	2006 - 2009		2009 - 2013		2014 - 2018		2019 - 2023	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-17.66***	-11.79***	-17.41***	-10.18***	-14.51***	-8.55***	-8.08***	-3.03***
	(0.36)	(0.67)	(0.31)	(0.56)	(0.28)	(0.53)	(0.34)	(0.64)
Immigration Age		-1.49***		-1.69***		-1.27***		-1.06***
		(0.14)		(0.11)		(0.10)		(0.12)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. Microdata Area	X	X	X	X	X	X	X	X
F.E. Sex	X	X	X	X	X	X	X	X
F.E. Education	X	X	X	X	X	X	X	X
F.E. Race	X	X	X	X	X	X	X	X
F.E. Hispanic	X	X	X	X	X	X	X	X
F.E. Survey Year	X	X	X	X	X	X	X	X
N	2,079,500	2,079,500	2,162,012	2,162,012	2,231,275	2,231,275	2,200,018	2,200,018
R^2	0.161	0.161	0.196	0.196	0.239	0.239	0.228	0.228
Mean Income, Immigration Age <21	$71,088	$71,088	$78,339	$78,339	$98,660	$98,660	$134,099	$134,099

Income, Controlling for Demographics, Employment & Occupation (ACS)

	Log (Income)							
	2006 - 2009		2009 - 2013		2014 - 2018		2019 - 2023	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Immigrant	-11.23***	-7.57***	-10.98***	-6.50***	-9.15***	-5.43***	-4.41***	-1.27**
	(0.36)	(0.65)	(0.30)	(0.54)	(0.27)	(0.51)	(0.33)	(0.61)
Immigration Age		-0.93***		-1.05***		-0.79***		-0.66***
		(0.14)		(0.11)		(0.09)		(0.11)
F.E. Birth Year	X	X	X	X	X	X	X	X
F.E. Microdata Area	X	X	X	X	X	X	X	X
F.E. Sex	X	X	X	X	X	X	X	X
F.E. Education	X	X	X	X	X	X	X	X
F.E. Race	X	X	X	X	X	X	X	X
F.E. Hispanic	X	X	X	X	X	X	X	X
F.E. Survey Year	X	X	X	X	X	X	X	X
F.E. Employment	X	X	X	X	X	X	X	X
F.E. Occupation	X	X	X	X	X	X	X	X
N	2,079,500	2,079,500	2,162,012	2,162,012	2,231,275	2,231,275	2,200,018	2,200,018
R^2	0.199	0.199	0.248	0.248	0.299	0.299	0.299	0.299
Mean Income, Immigration Age <21	\$71,088	\$71,088	\$78,339	\$78,339	\$98,660	\$98,660	\$134,099	\$134,099

Differences in Mortgage & Homeownership Remain After Controlling for Income & Demographics (5-year ACS waves)



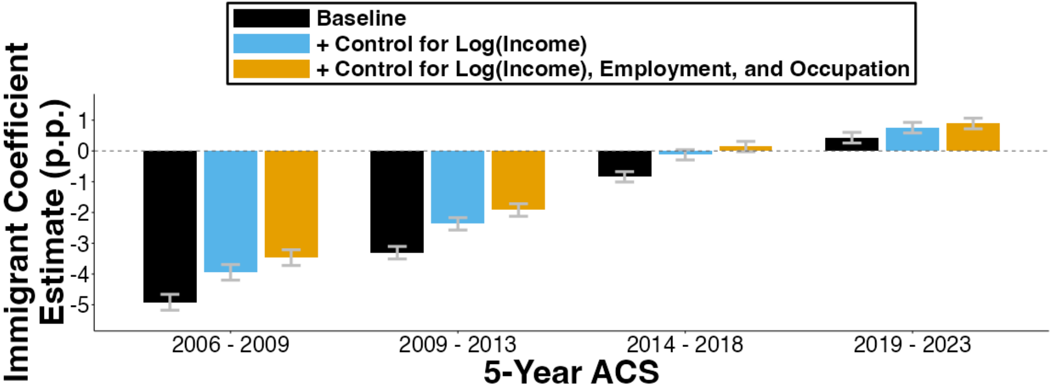


Immigrating one year later in life leads to persistently lower mortgage use & homeownership

All regressions include fixed effects for...

birth year, geographic area, sex, education levels, race, hispanic, & survey year

Car Ownership Difference Disappears (5-year ACS waves)



All regressions include fixed effects for...

birth year, geographic area, sex, education levels, race, hispanic, & survey year

Access to mortgage, homeownership, and car ownership (ACS)

	2006 - 2009			2009 - 2013			2014 - 2018			2019 - 2023		
A. Dep Var: Any Mortgage	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-9.90***	-7.57***	-6.90***	-8.09***	-6.21***	-5.38***	-3.56***	-2.09***	-1.23***	-1.86***	-1.40***	-0.65*
	(0.31)	(0.30)	(0.30)	(0.28)	(0.27)	(0.27)	(0.29)	(0.28)	(0.28)	(0.35)	(0.34)	(0.34)
Immigration Age	-1.98***	-1.69***	-1.73***	-1.63***	-1.32***	-1.34***	-1.36***	-1.14***	-1.13***	-0.76***	-0.60***	-0.56***
	(0.06)	(0.06)	(0.06)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.06)	(0.06)	(0.06)
Log (Income)		17.92***	18.13***		16.43***	16.41***		15.23***	14.88***		13.88***	13.12***
		(0.05)	(0.05)		(0.05)	(0.05)		(0.05)	(0.05)		(0.06)	(0.06)
R^2	0.100	0.213	0.220	0.116	0.205	0.212	0.135	0.206	0.213	0.131	0.192	0.199
Mean, Immigration Age <21	44.22	44.22	44.22	44.65	44.65	44.65	48.12	48.12	48.12	54.25	54.25	54.25

B. Dep Var: Homeowner	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-13.08***	-10.56***	-10.19***	-10.65***	-8.67***	-8.04***	-4.82***	-3.31***	-2.50***	-0.83**	-0.37	0.35
	(0.32)	(0.32)	(0.32)	(0.30)	(0.29)	(0.29)	(0.30)	(0.29)	(0.29)	(0.35)	(0.34)	(0.34)
Immigration Age	-2.02***	-1.70***	-1.79***	-1.75***	-1.42***	-1.50***	-1.56***	-1.34***	-1.38***	-1.10***	-0.93***	-0.94***
	(0.07)	(0.07)	(0.06)	(0.06)	(0.06)	(0.06)	(0.05)	(0.05)	(0.05)	(0.06)	(0.06)	(0.06)
Log (Income)		19.37***	19.91***		17.31***	17.78***		15.62***	15.86***		13.80***	13.71***
		(0.05)	(0.05)		(0.05)	(0.05)		(0.05)	(0.05)		(0.06)	(0.06)
R^2	0.114	0.244	0.255	0.128	0.225	0.235	0.152	0.229	0.237	0.155	0.224	0.233
Mean, Immigration Age <21	52.40	52.40	52.40	53.25	53.25	53.25	59.26	59.26	59.26	68.54	68.54	68.54

C. Dep Var: Any Car	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Immigrant	-2.37***	-1.72***	-1.34***	-0.88***	-0.34*	0.00	1.11***	1.53***	1.69***	1.18***	1.30***	1.39***
	(0.25)	(0.24)	(0.24)	(0.20)	(0.19)	(0.19)	(0.16)	(0.16)	(0.16)	(0.17)	(0.17)	(0.17)
Immigration Age	-0.65***	-0.56***	-0.54***	-0.57***	-0.48***	-0.45***	-0.41***	-0.35***	-0.33***	-0.16***	-0.11***	-0.10***
	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Log (Income)		4.99***	4.85***		4.80***	4.61***		4.36***	4.15***		3.67***	3.48***
		(0.03)	(0.03)		(0.03)	(0.03)		(0.03)	(0.03)		(0.03)	(0.03)
R^2	0.200	0.234	0.236	0.211	0.240	0.243	0.187	0.215	0.219	0.153	0.177	0.181
Mean, Immigration Age <21	93.51	93.51	93.51	93.66	93.66	93.66	94.69	94.69	94.69	95.40	95.40	95.40